

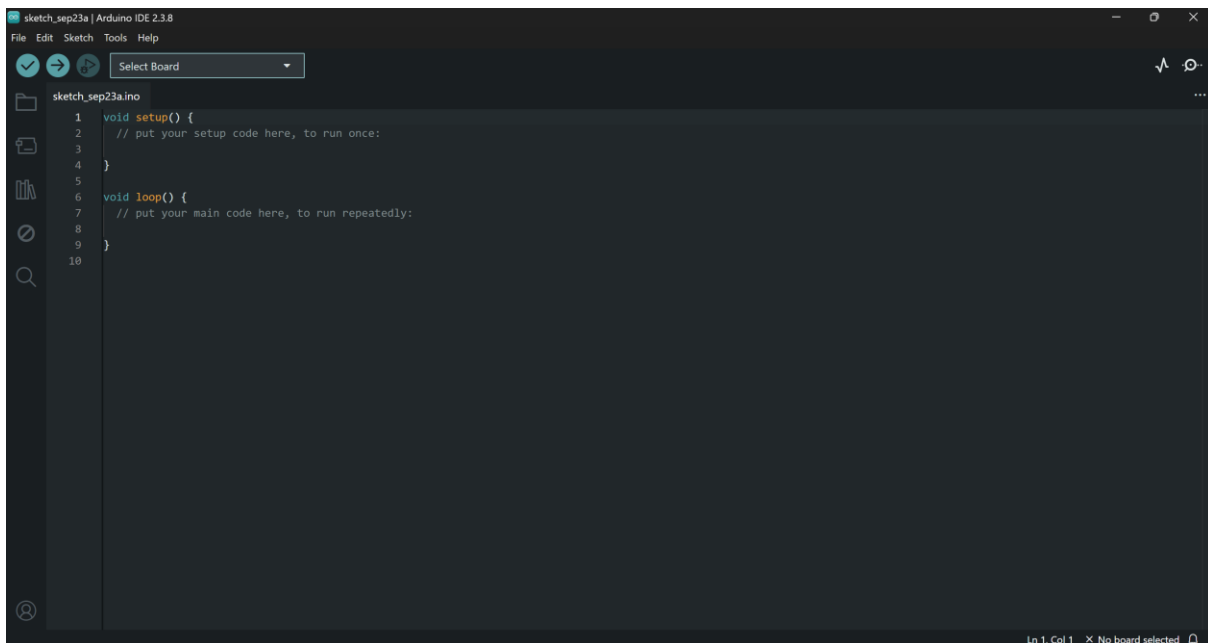
Laboratorium Dasar Teknik Elektro – STEI ITB



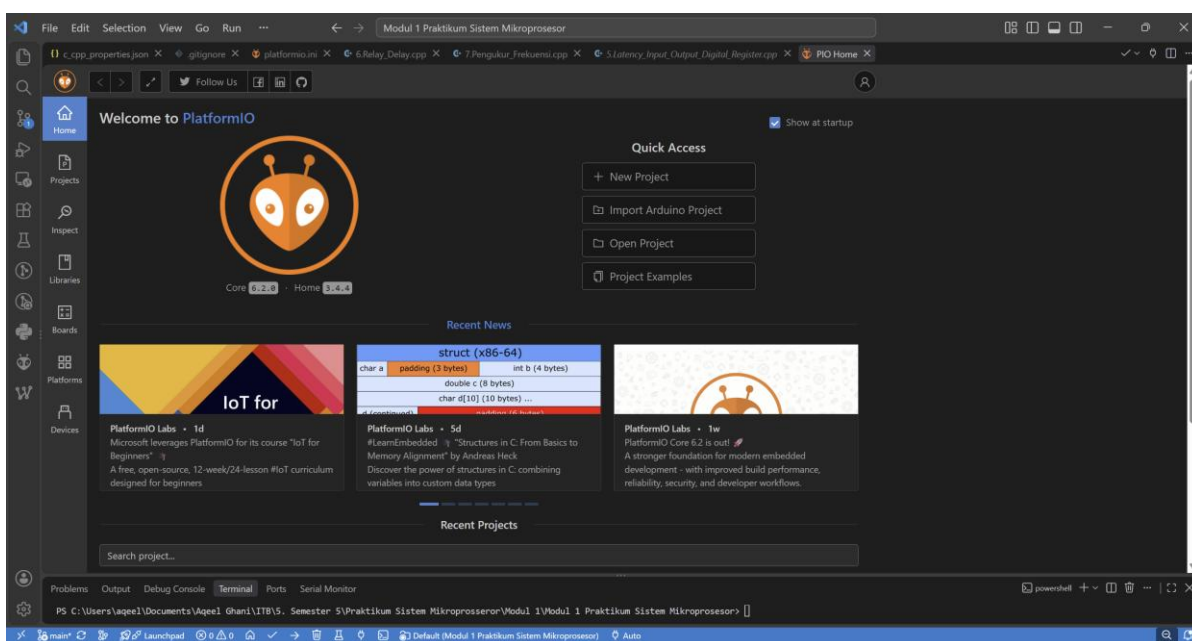
Nama Kuliah (Kode) : Praktikum Sistem Mikroprosesor (EL3112)
Tahun / Semester : 2026-2027 / Ganjil
Modul : 1
Nama Asisten / NIM :
Nama Praktikan/NIM : Muhammad Aqeel Ghani / 13224071

Tugas Pendahuluan

1. Instalasi Arduino IDE



2. Instalasi PlatformIO di Visual Studio Code



3. Mencoba Compile contoh-contoh kode yang diberikan

3.1. Output Digital dengan akses register ESP32

```
#include "soc/gpio_reg.h"
#include "driver/gpio.h"
#include <Arduino.h>

#define LED_PIN 2 // Built-in LED on most ESP32 boards
#define BLINK_DELAY_MS 500 // 500ms on + 500ms off = 1Hz

void setup() {
    // Configure GPIO pin as output using register access
    // Enable the GPIO pin in the output enable register
    REG_WRITE(GPIO_ENABLE_REG, REG_READ(GPIO_ENABLE_REG) | (1 << LED_PIN));

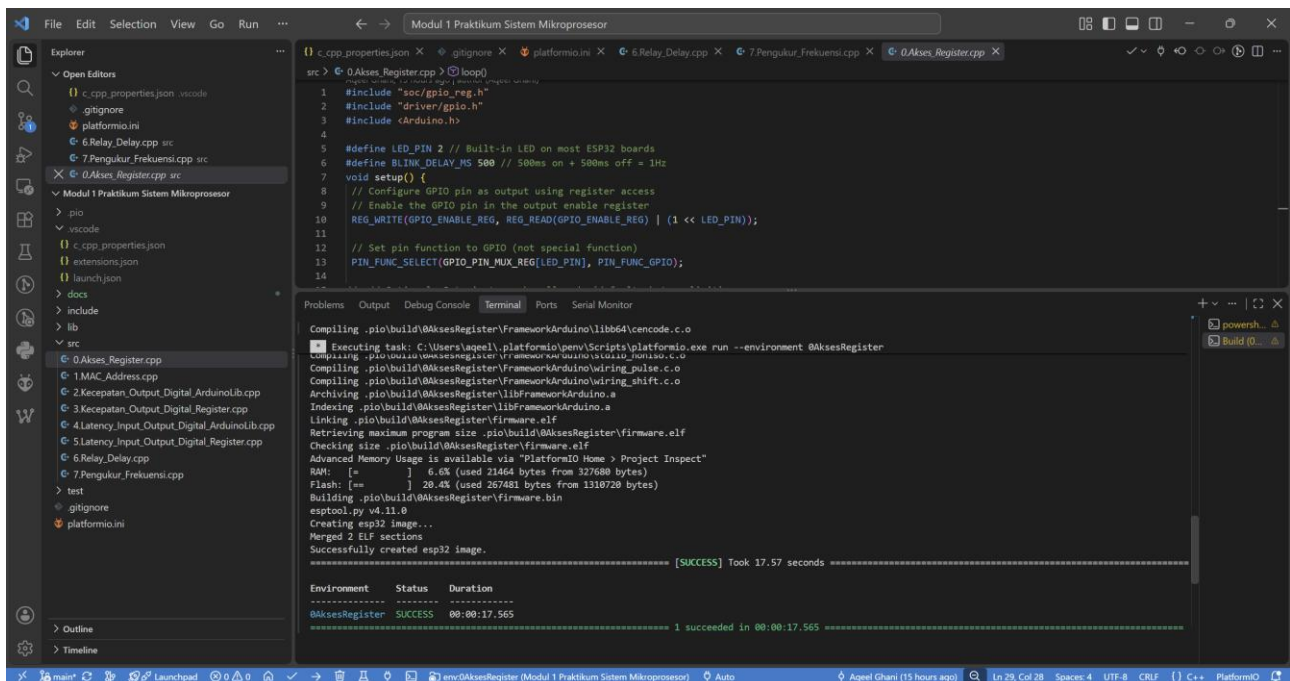
    // Set pin function to GPIO (not special function)
    PIN_FUNC_SELECT(GPIO_PIN_MUX_REG[LED_PIN], PIN_FUNC_GPIO);

    // // Optional: Set pin to push-pull mode (default, but explicit)
    // REG_SET_BIT(GPIO_PIN_REG[LED_PIN], GPIO_PIN_PAD_DRIVER);

    Serial.begin(115200);
    Serial.println("ESP32 Register-based LED Blink Started");
}

void loop() {
    // Turn LED ON - Set the bit high using register access
    REG_WRITE(GPIO_OUT_REG, REG_READ(GPIO_OUT_REG) | (1 << LED_PIN));
    Serial.println("LED ON");
    delay(BLINK_DELAY_MS);

    // Turn LED OFF - Clear the bit using register access
    REG_WRITE(GPIO_OUT_REG, REG_READ(GPIO_OUT_REG) & ~(1 << LED_PIN));
    Serial.println("LED OFF");
    delay(BLINK_DELAY_MS);
}
```



3.2. Mencatat MAC Address (Identitas) ESP32

```
#include <WiFi.h>
#include <esp_wifi.h>

void readMacAddress() {
    uint8_t baseMac[6];
    esp_err_t ret = esp_wifi_get_mac(WIFI_IF_STA, baseMac);
}
```

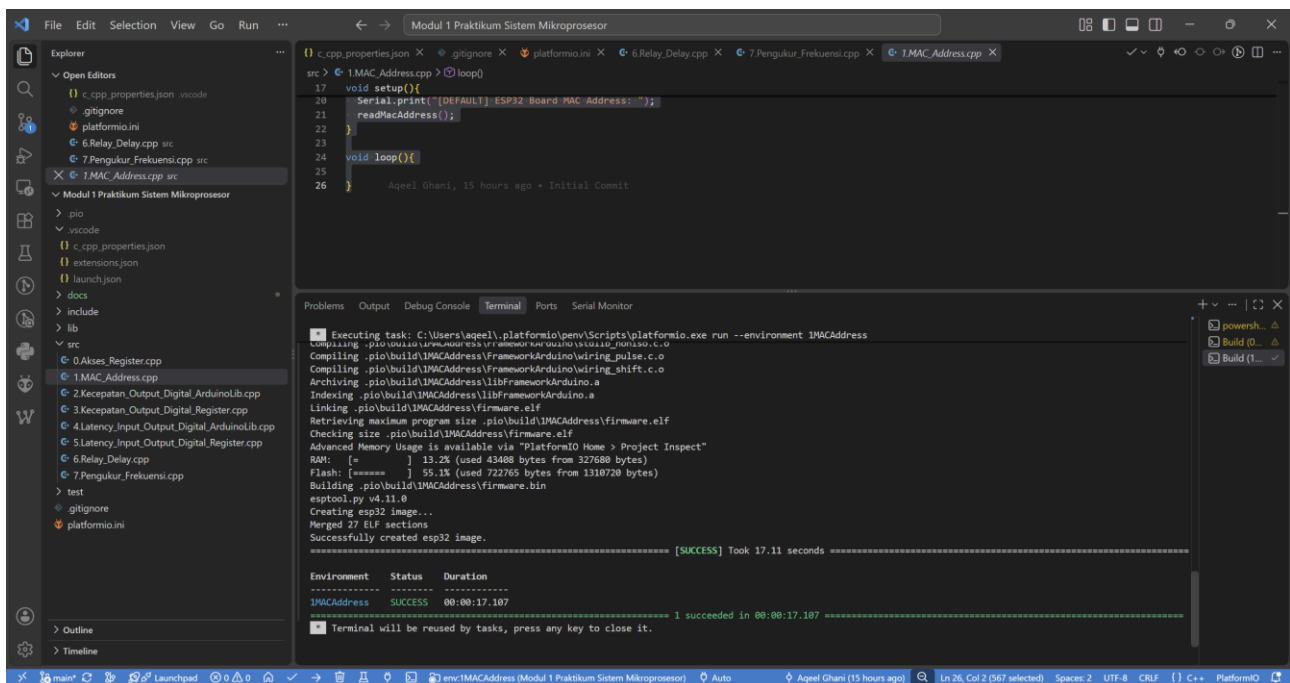
```

if (ret == ESP_OK) {
    Serial.printf("%02x:%02x:%02x:%02x:%02x:%02x\n",
        baseMac[0], baseMac[1], baseMac[2],
        baseMac[3], baseMac[4], baseMac[5]);
} else {
    Serial.println("Failed to read MAC address");
}
}

void setup() {
    Serial.begin(115200);
    WiFi.mode(WIFI_STA);
    Serial.print("[DEFAULT] ESP32 Board MAC Address: ");
    readMacAddress();
}

void loop() {
}

```



3.3. Mengukur Kecepatan Output Digital dengan Library Arduino

```

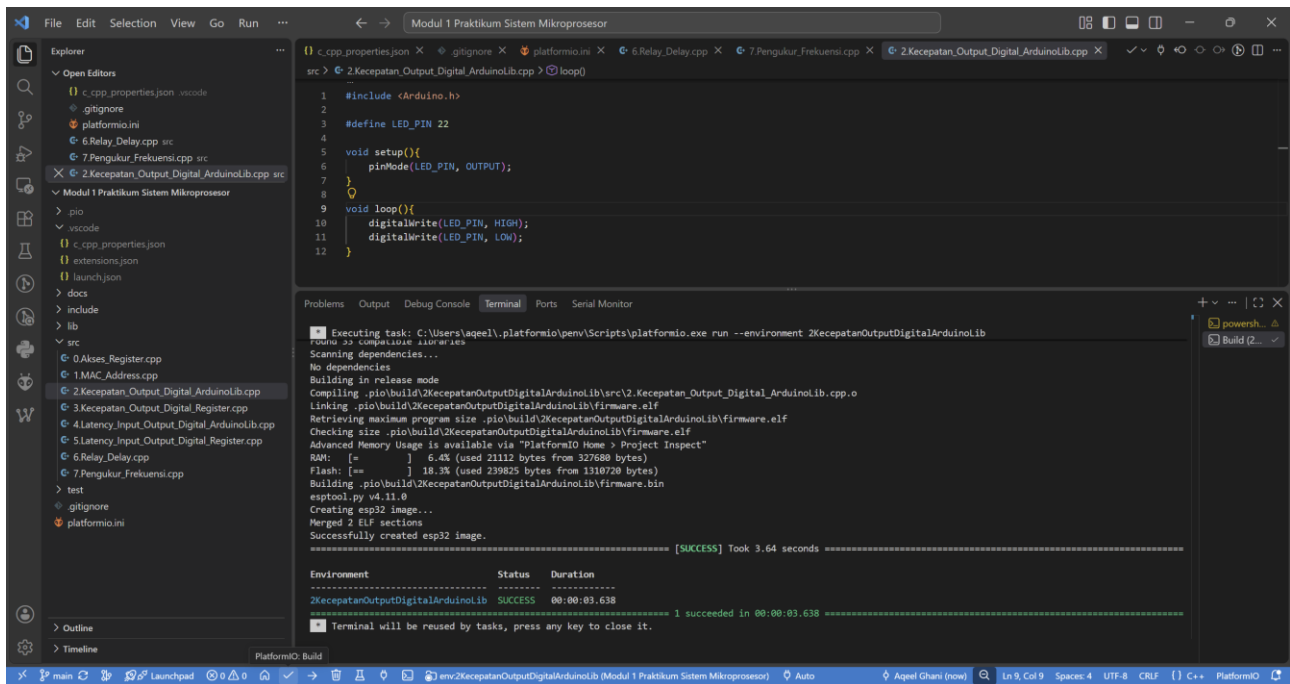
#include <Arduino.h>

#define LED_PIN 22

void setup() {
    pinMode(LED_PIN, OUTPUT);
}

void loop() {
    digitalWrite(LED_PIN, HIGH);
    digitalWrite(LED_PIN, LOW);
}

```



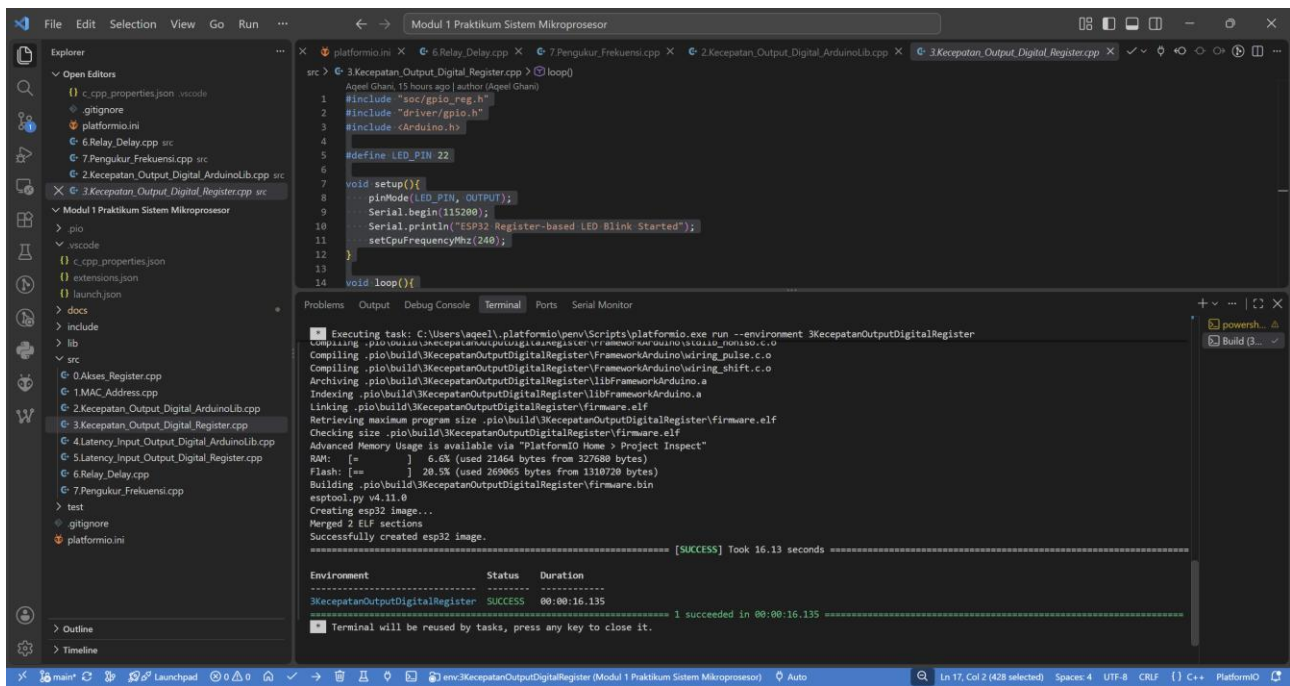
3.4. Mengukur Kecepatan Output Digital dengan Akses Register

```
#include "soc/gpio_reg.h"
#include "driver/gpio.h"
#include <Arduino.h>

#define LED_PIN 22

void setup() {
    pinMode(LED_PIN, OUTPUT);
    Serial.begin(115200);
    Serial.println("ESP32 Register-based LED Blink Started");
    setCpuFrequencyMhz(240);
}

void loop() {
    REG_WRITE(GPIO_OUT_REG, REG_READ(GPIO_OUT_REG) | (1 << LED_PIN));
    REG_WRITE(GPIO_OUT_REG, REG_READ(GPIO_OUT_REG) & ~(1 << LED_PIN));
}
```



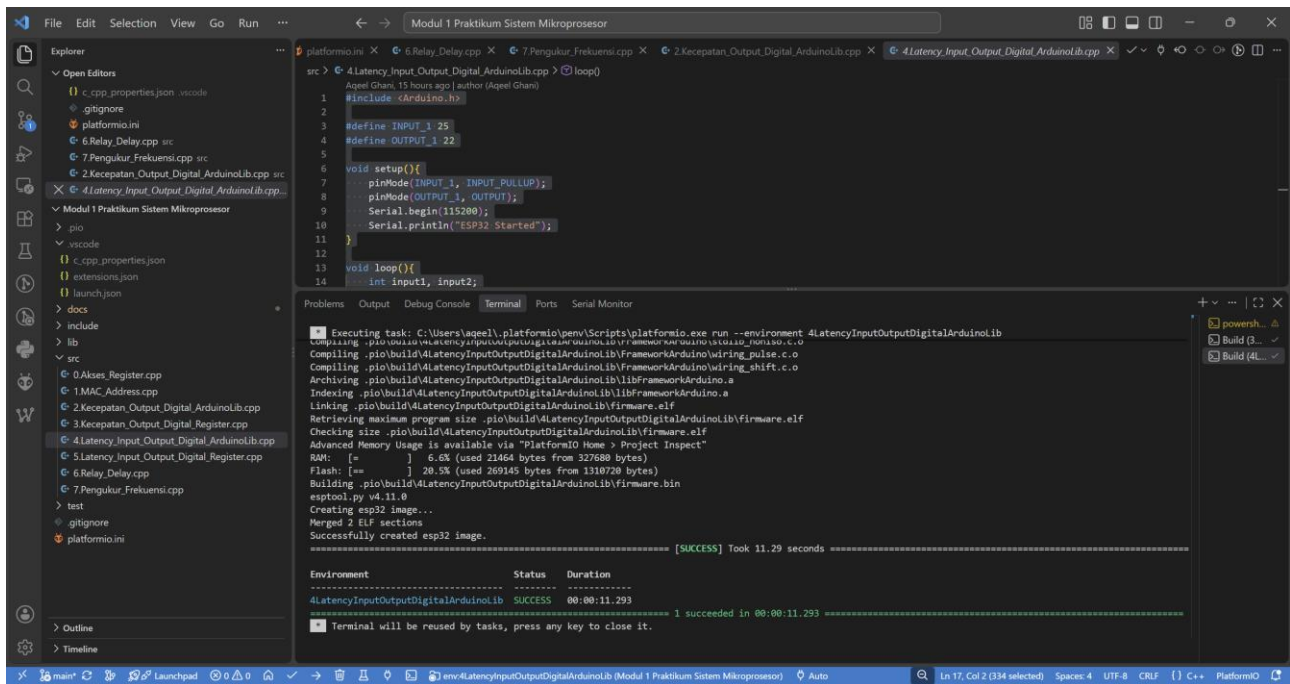
3.5. Latency Input Output Digital dengan Library Arduino

```
#include <Arduino.h>

#define INPUT_1 25
#define OUTPUT_1 22

void setup() {
    pinMode(INPUT_1, INPUT_PULLUP);
    pinMode(OUTPUT_1, OUTPUT);
    Serial.begin(115200);
    Serial.println("ESP32 Started");
}

void loop() {
    int input1, input2;
    input1 = digitalRead(INPUT_1);
    digitalWrite(OUTPUT_1, input1);
}
```



4. Mempersiapkan tugas-tugas kode yang diberikan

4.1. Latency Input Output Digital dengan Akses Register

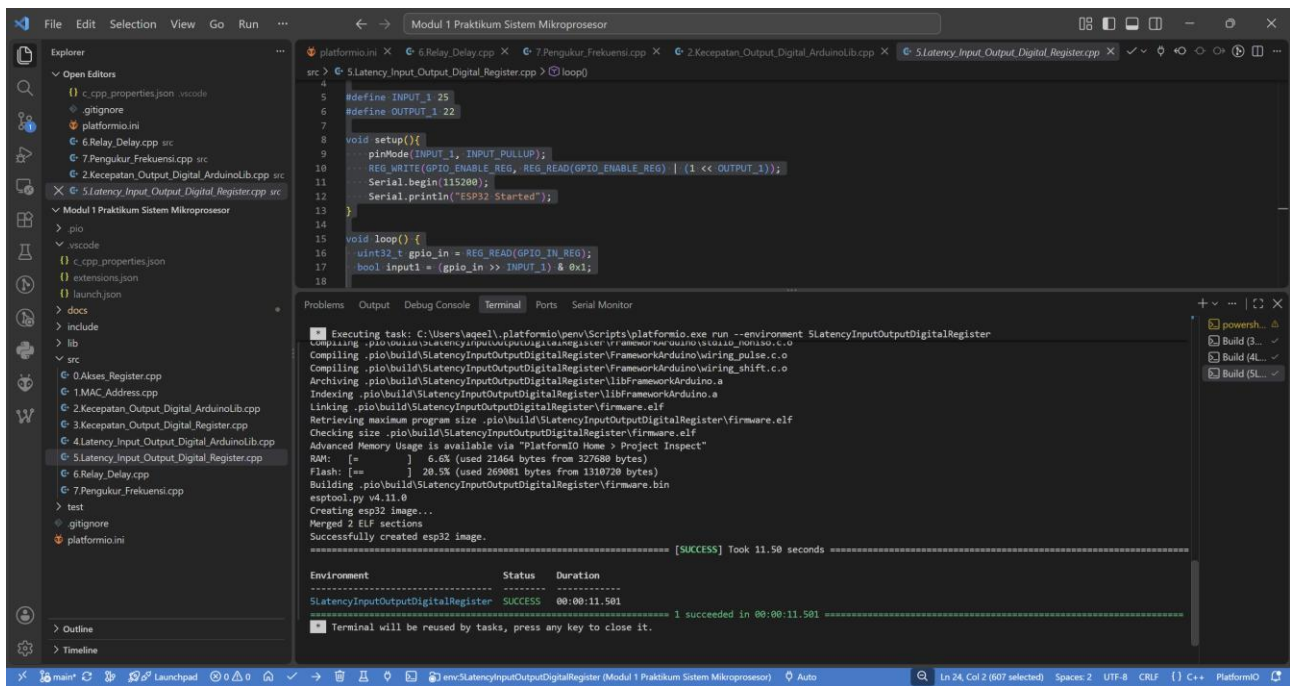
```
#include <Arduino.h>
#include "soc/gpio_reg.h"
#include "driver/gpio.h"

#define INPUT_1 25
#define OUTPUT_1 22

void setup(){
    pinMode(INPUT_1, INPUT_PULLUP);
    REG_WRITE(GPIO_ENABLE_REG, REG_READ(GPIO_ENABLE_REG) | (1 << OUTPUT_1));
    Serial.begin(115200);
    Serial.println("ESP32 Started");
}

void loop() {
    uint32_t gpio_in = REG_READ(GPIO_IN_REG);
    bool input1 = (gpio_in >> INPUT_1) & 0x1;

    if (input1) {
        REG_WRITE(GPIO_OUT_REG, REG_READ(GPIO_OUT_REG) | (1 << OUTPUT_1));
    } else {
        REG_WRITE(GPIO_OUT_REG, REG_READ(GPIO_OUT_REG) & ~(1 << OUTPUT_1));
    }
}
```



4.2. Pengendali Relay

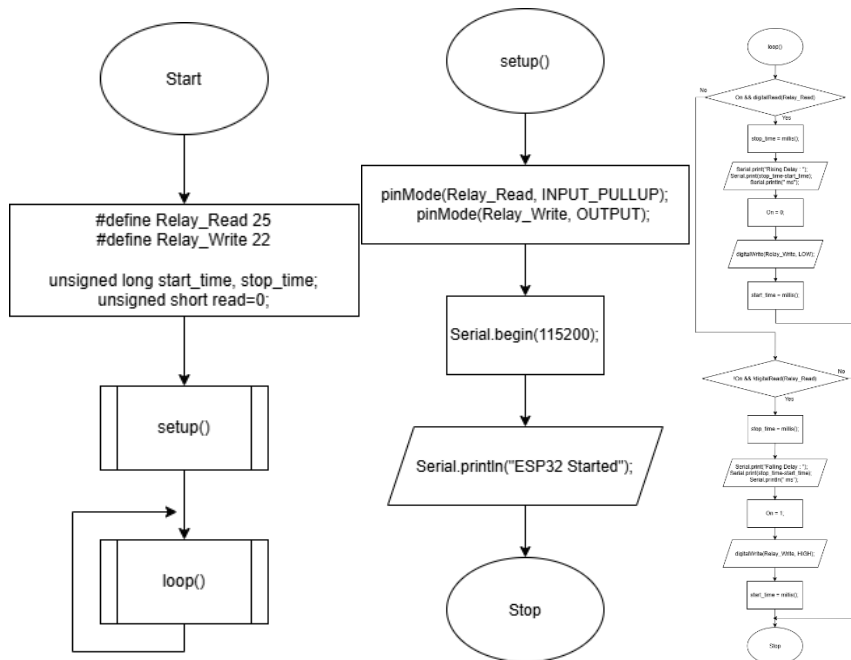
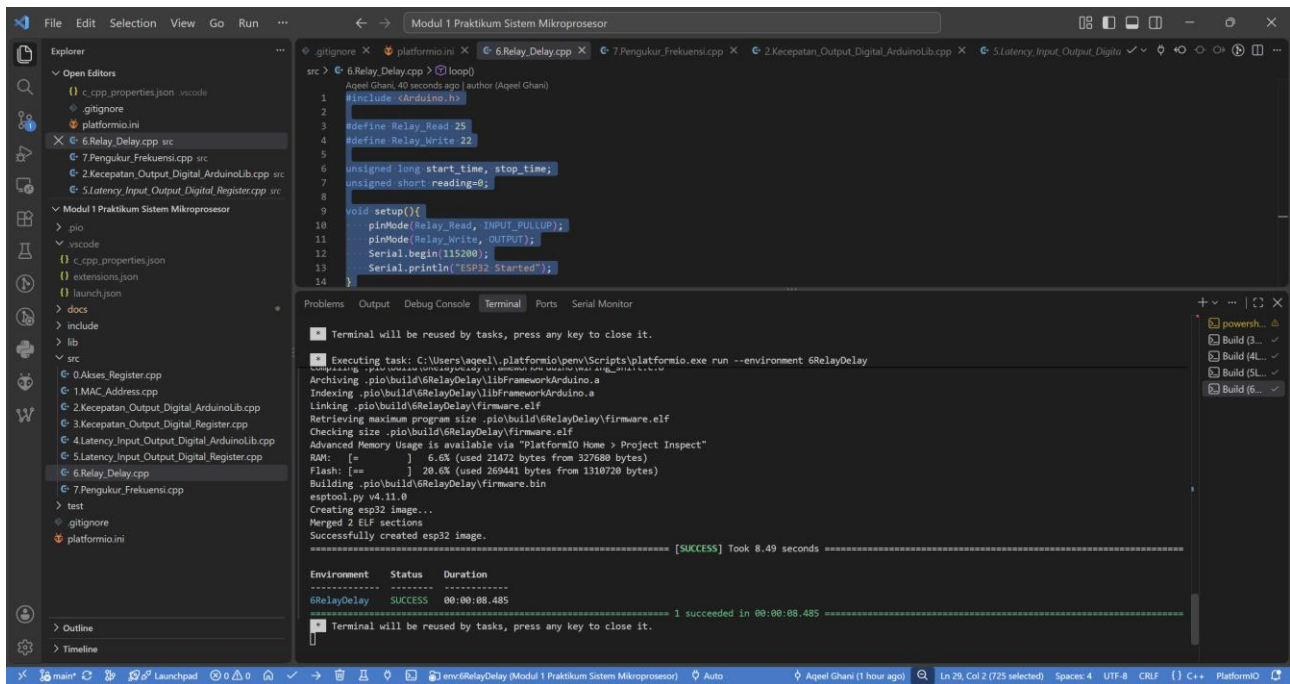
```
#include <Arduino.h>

#define Relay_Read 25
#define Relay_Write 22

unsigned long start_time = 0 , stop_time = 0;
unsigned short On=0;

void setup(){
    pinMode(Relay_Read, INPUT_PULLUP);
    pinMode(Relay_Write, OUTPUT);
    Serial.begin(115200);
    Serial.println("ESP32 Started");
}

void loop(){
    if (On && digitalRead(Relay_Read)){
        stop_time = millis();
        Serial.print("Rising Delay : ");
        Serial.print(stop_time-start_time);
        Serial.println(" ms");
        On = 0;
        digitalWrite(Relay_Write, LOW);
        start_time = millis();
    } else if (!On && !digitalRead(Relay_Read)){
        stop_time = millis();
        Serial.print("Falling Delay : ");
        Serial.print(stop_time-start_time);
        Serial.println(" ms");
        On = 1;
        digitalWrite(Relay_Write, HIGH);
        start_time = millis();
    }
}
```



4.3. Pengukur Frekuensi

```

#include <Arduino.h>

#define input1 25

int counter = 0;

void IRAM_ATTR count_interup();

void setup(){
  pinMode(input1, INPUT);
  Serial.begin(115200);
  Serial.println("ESP32 Started");
  attachInterrupt(digitalPinToInterrupt(input1), count_interup, RISING);
}

void loop(){
  delay(1000);
  
```

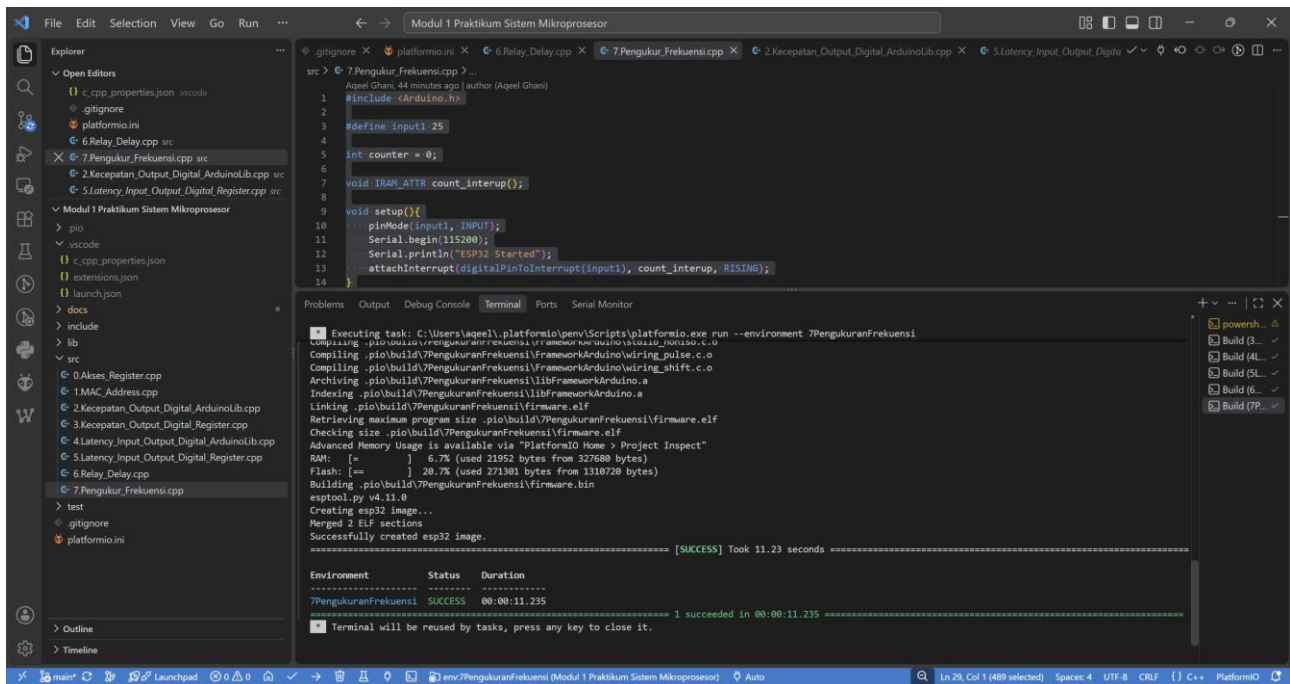
```

Serial.print("Frekuensi ");
Serial.print(counter);
Serial.println("Hz");

counter = 0;
}

void IRAM_ATTR count_interup() {
    counter++;
}

```



5. Menyiapkan tabel-tabel pengukuran di setiap percobaan

